

6517B

Electrometer/High Resistance Meter

Referenced
in 1000s of
research papers



- Measures resistances up to $10^{18}\Omega$
- 10aA ($10 \times 10^{-18}\text{A}$) current measurement resolution
- $<3\text{fA}$ input bias current
- $6\frac{1}{2}$ -digit high accuracy measurement mode
- $<20\mu\text{V}$ burden voltage on the lowest current ranges
- Voltage measurements up to 200V with $>200\text{T}\Omega$ input impedance
- Built-in $\pm 1000\text{V}$ voltage source
- Unique alternating polarity voltage sourcing and measurement method for high resistance measurements
- Built-in test sequences for four different device characterization tests, surface and volume resistivity, surface insulation resistance, and voltage sweeping
- Optional plug-in scanner cards for testing up to 10 devices or material samples with one test setup

APPLICATIONS

- Polymer electrical characterization
- Beam measurements
- Dosimetry
- Device leakage current measurements
- Insulation resistance measurements
- Optoelectronic detector characterization
- Volume and surface resistivity
- Nanomaterial characterization

The Keithley 6517B Electrometer/High Resistance Meter is the worldwide research laboratory standard for sensitive measurements. With over 60 years of low level measurement expertise, Keithley electrometers provide reliable measurements of current levels down to 10aA ($10 \times 10^{-18}\text{A}$), charge levels down to 1fC, and the highest resistance measurements available up to $10^{18}\Omega$. The 6517B is also capable of measuring the largest voltage range—up to 200V—with an input impedance exceeding $200\text{T}\Omega$.

Exceptional Performance Specifications

The 6517B has incorporated Keithley's decades of expertise in low level measurement technology into an innovative, low current input amplifier with an input bias current of $<3\text{fA}$, just 0.75fA/p-p noise, and $<20\mu\text{A}$ burden voltage on the low-

est current ranges. The voltage circuit input impedance is greater than $200\text{T}\Omega$ for near-ideal circuit loading. These specifications ensure the accuracy and sensitivity needed for accurate low current and high impedance voltage, resistance, and charge measurements in areas of research such as physics, optics, nanotechnology, and materials science. A built-in $\pm 1\text{kV}$ voltage source with sweep capability simplifies performing leakage, breakdown, and resistance testing, as well as volume ($\Omega\text{-cm}$) and surface resistivity (Ω/square) measurements on insulating materials.

Wide Measurement Ranges

The 6517B offers autoranging over the full span of ranges on current, resistance, voltage, and charge measurements. The 6517B combines the following measurement capabilities:

- Ultra-sensitive ammeter with current measurement from 10aA to 20mA
- Highest impedance voltmeter with voltage measurement from $1\mu\text{V}$ to 200V
- Ultra-high range ohmmeter with resistance measurement from 1Ω to $10^{18}\Omega$
- Sensitive coulombmeter with charge measurement from 1fC to $2\mu\text{C}$

Improved High Resistivity Measurements

Many test applications require measuring high levels of resistivity (surface or volume) of materials. The conventional method of making these measurements is to apply a sufficiently large voltage to a sample, measure the current that flows through the sample, then calculate the resistance using Ohm's Law ($R=V/I$). While high resistance materials and devices produce very small currents that are difficult to measure accurately, Keithley electrometers and picoammeters are used successfully for such measurements.

Even with high quality instrumentation, inherent background currents in the material can make these measurements difficult to perform accurately. Insulating materials, polymers, and plastics typically exhibit background currents due to piezoelectric effects, capacitive elements charged by static electricity, and polarization effects. These background currents are often equal to or greater than the current stimulated by the applied voltage. In these cases, the result is often unstable, providing inaccurate resistance or resistivity readings or even erroneous negative values. Keithley's 6517B is designed to solve these problems and provides consistent, repeatable, and accurate measurements for a wide variety of materials and components, especially when used in combination with the 8009 Resistivity Test Fixture.

Alternating Polarity Method for High Resistance Measurements

The 6517B uses the Alternating Polarity Method, which virtually eliminates the effect of any background currents in the sample. First and second order drifts of the background currents are also canceled out. The Alternating Polarity Method applies a voltage of positive polarity, then the current is measured after a specified delay (Measure Time). Next, the polarity is reversed and the current measured again, using the same delay. This process is repeated continuously, and the resistance is calculated based on a weighted average of the four most recent current measurements. This method typically produces a highly repeatable, accurate measurement of resistance (or resistivity) by the seventh rever-

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Ordering Information

6517B Electrometer/High Resistance Meter

Accessories Supplied

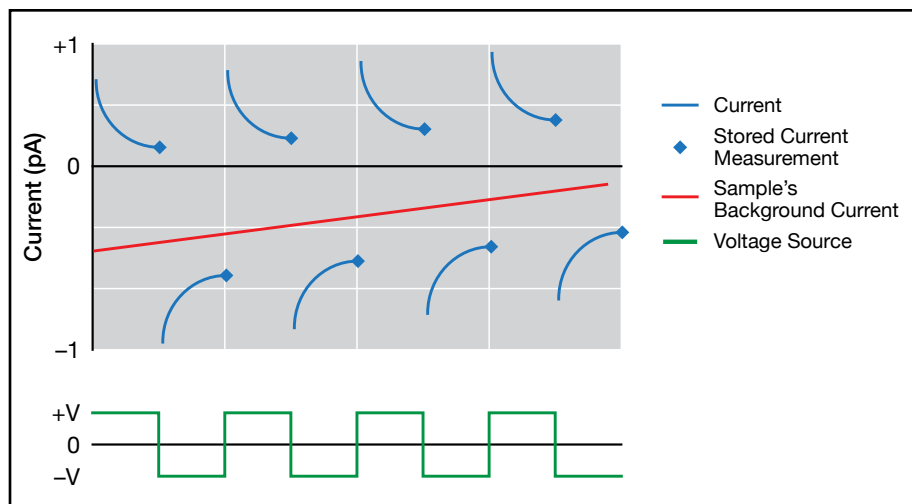
237-ALG-2 Low Noise Triax Cable, 3-slot Triax to Alligator Clips, 2m (6.6 ft)

8607 Safety High Voltage Dual Test Leads

6517-TP Thermocouple Bead Probe

CS-1305 Interlock Connector

Electrometer/High Resistance Meter



The alternating voltage source polarity method eliminates the effects of background currents in materials for making repeatable, accurate high resistance and resistivity measurements.

sal on most materials (i.e., by discarding the first three readings.) For example, a 1mm-thick sample of $10^{14}\Omega\text{-cm}$ material can be measured with 0.3% repeatability in the 8009 Test Fixture, provided the background current changes less than 200fA over a 15-second period.

Simple DMM-like Operation

The 6517B is designed for easy, DMM-like operation via the front panel, with single-button control of important functions such as resistance measurement. Also, all the functions of the 6517B can be programmed through either the RS-232 interface or the GPIB (IEEE-488) interface.

High Accuracy High Resistance Measurements

The 6517B offers a number of features and capabilities that help ensure the accuracy of high resistance measurement applications. For example, the built-in voltage source simplifies determining the relationship between an insulator's resistivity and the level of source voltage used. It is well-suited for capacitor leakage and insulation resistance measurements, tests of the surface insulation resistance of printed circuit boards, voltage coefficient testing of resistors, and diode leakage characterization.

Temperature and Humidity Stamping

Humidity and temperature can influence the resistivity values of materials significantly. To help you make accurate comparisons of readings acquired under varying conditions, the 6517B offers a built-in type K thermocouple and an optional 6517-RH Relative Humidity Probe. A built-in 50,000 reading data storage buffer allows recording and recalling measurements stamped with the time of the measurement, the temperature, and the relative humidity.

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Internal Test Sequences Expand and Simplify Applications

The 6517B has a number of internal test sequences that assists in easily setting up and performing a number of tests. Device characterization sequences include diode leakage current measurement, capacitor leakage current measurement, cable insulation resistance measurement, and resistor voltage coefficient measurement. Resistivity and resistance tests include volume resistivity, surface resistivity, and surface insulation resistance testing. Parameters can be characterized as a function of voltage with the square wave and staircase test sequences.

In addition to its built-in tests, the 6517B excels in low current, high impedance voltage, resistance, and charge measurements in areas of research such as physics, optics, and materials science. The electrometer's extremely low voltage burden makes it particularly valuable for use in solar cell characterization applications and its built-in voltage source and low current sensitivity make it an excellent solution for high resistance measurements of nanomaterials such as polymer-based nanowires, other nanomaterials, ceramics, dielectric films, and biomaterials.

With its highly responsive measurements and DMM-like operation, the 6517B performs well in quality control, design engineering, and production test applications involving leakage current, breakdown, and resistance testing. Volume and surface resistivity measurements on non-conductive materials are particularly enhanced by the 6517B's voltage reversal method. The 6517B is also excellent for electrochemistry applications such as high impedance, ion-selective electrodes and pH measurements, conductivity cells, and potentiometry.

SERVICES AVAILABLE

6517B-EW	1 Year KeithleyCare Gold Extended Warranty Plan
6517B-3Y-EW-STD	3 Year KeithleyCare Gold Extended Warranty Plan
6517B-5Y-EW-STD	5 Year KeithleyCare Gold Extended Warranty Plan
C/6517B-3Y-STD	KeithleyCare 3-Calibration, 3-Year Standard Calibration Plan
C/6517B-3Y-DATA	KeithleyCare 3-Calibration, 3-Year Calibration Plan with Data
C/6517B-3Y-17025	KeithleyCare 3-Calibration, 3-Year ISO 17025 Calibration Plan
C/6517B-5Y-STD	KeithleyCare 5-Calibration, 5-Year Standard Calibration Plan
C/6517B-5Y-DATA	KeithleyCare 5-Calibration, 5-Year Calibration Plan with Data
C/6517B-5Y-17025	KeithleyCare 5-Calibration, 5-Year ISO 17025 Calibration Plan

ACCESSORIES AVAILABLE

CABLES

6517B-ILC-3	Interlock Cable
7007-1	Shielded IEEE-488 Cable, 1m (3.2 ft)
7007-2	Shielded IEEE-488 Cable, 2m (6.5 ft)
7009-5	RS-232 Cable
7078-TRX-x	Low Noise Triax Cable, 3-Slot Triax Connectors, x=3: 0.9m (3 ft), x=10: 3m (10 ft), x=20: 6m (20 ft)
8501-1	Trigger Link Cable, 1m (3.3 ft)
8501-2	Trigger Link Cable, 2m (6.6 ft)
8503	Trigger Link Cable to 2 male BNCs, 1m (3.3 ft)
8607	1kV Source Banana Cables

PROBES

6517-RH	Humidity Probe with Extension Cable
6517-TP	Temperature Bead Probe (included with 6517B)

TEST FIXTURE

8009	Resistivity Test Fixture
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OTHER

CS-1305	Interlock Connector
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ADAPTERS

237-BNC-TRX	Male BNC to 3-Lug Female Triax Adapter
237-TRX-NG	Triax Male-Female Adapter with Guard Disconnected
237-TRX-T	3-Slot Male Triax to Dual 3-Lug Female Triax Tee Adapter
237-TRX-TBC	3-Lug Female Triax Bulkhead Connector (1.1kV rated)
7078-TRX-BNC	3-Slot Male Triax to BNC Adapter
7078-TRX-GND	3-Slot Male Triax to BNC Adapter, guard removed
7078-TRX-TBC	3-Lug Female Triax Bulkhead Connector with Cap

RACK MOUNT KITS

4288-1	Single Fixed Rack Mounting Kit
4288-2	Dual Fixed Rack Mounting Kit
4288-4	Shelf Rack Mount kit, for 3U and 2U high instruments
4288-5	Shelf Mount Rack Kit, for two 2U high instruments
4299-7	Universal Shelf Mount Rack Kit

SCANNER CARDS

6521	Low Current Scanner Card
6522	Voltage/Low Current Scanner Card

GPIB INTERFACES

KPCI-488LPA	IEEE-488 Interface/Controller for the PCI Bus
KUSB-488B	IEEE-488 USB-to-GPIB Interface Adapter

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Accessories Extend Measurement Capabilities

A variety of optional accessories can be used to extend the 6517B applications and enhance its performance.

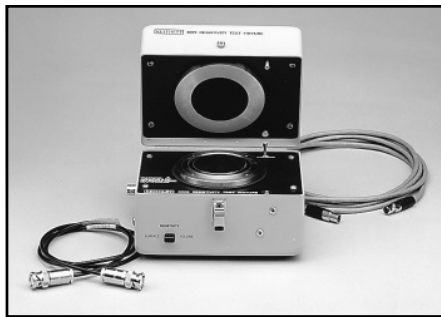
The 8009 Resistivity Chamber is a guarded test fixture for measuring volume and surface resistivities of sample materials. It has stainless-steel electrodes built to ASTM standards. The fixture's electrode dimensions are pre-programmed into the 6517B, so there's no need to calculate those values then enter them manually. This accessory is designed to protect you from contact with potentially hazardous voltages—opening the lid of the chamber automatically turns off the 6517B's voltage source.

6521 and 6522 Low Current, 10-Channel Scanner Cards

Two optional 10-channel plug-in scanner cards are available to extend the measurement performance of the 6517B Electrometer/High Resistance Meter. The cards install directly into the option slot in the back panel of the 6517B. The cards are also compatible with the 6517A and 6517.

The 6521 Low Current Scanner Card is a 10-channel multiplexer, designed for switching low currents in multipoint testing applications or when the test configuration must be changed. Offset current on each channel is $<1\text{pA}$ and high isolation is maintained between each channel ($>10^{13}\Omega$). The 6521 maintains the current path even when the channel is deselected, making it a true current switch. BNC input connectors help provide shielding for sensitive measurements and make the card compatible with low noise coaxial cables. The 6521 is well suited for automating reverse leakage tests on semiconductor junctions or gate leakage tests on FETs.

The 6522 Voltage/Low Current Scanner Card can provide up to ten channels of low-level current, high impedance voltage, high resistance, or charge switching. Although it's similar to the 6521 in many ways, the 6522's input connectors are 3-lug triax. The card can be software configured for high impedance voltage switching of up to 200V. Triaxial connectors make it possible to float the card 500V above ground and drive guard to 200V.

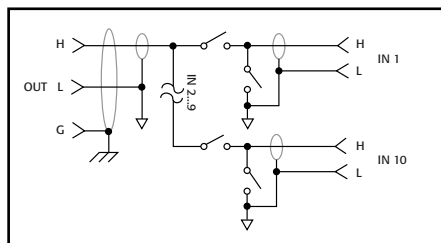


8009 Resistivity Chamber is compliant with American Society for Testing and Materials (ASTM) Standard D 257 Standard Test Methods for DC Resistance or Conductance of Insulating Materials. The 8009 combined with the 6517B provides a complete system for making high quality, safe resistivity measurements.



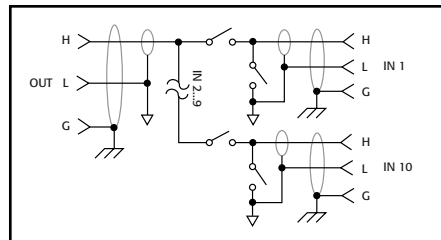
6521 SPECIFICATIONS

CHANNELS PER CARD: 10.
FUNCTIONS: Amps.
CONTACT CONFIGURATION: Single pole, "break-before-make" for signal HI input. Signal LO is common for all 10 channels and output. When a channel is off, signal HI is connected to signal LO.
CONNECTOR TYPE: Inputs BNC, Outputs Triaxial.
SIGNAL LEVEL: 30V, 500mA, 10VA (resistive load).
CONTACT LIFE: $>10^6$ closures at maximum signal level; $>10^7$ closures at low signal levels.
CONTACT RESISTANCE: $<1\Omega$.
CONTACT POTENTIAL: $<200\mu\text{V}$.
OFFSET CURRENT: $<1\text{pA}$ ($<30\text{fA}$ typical at 23°C , $<60\%$ RH).
ACTUATION TIME: 2ms.
COMMON MODE VOLTAGE: $<30\text{V}$ peak.
ENVIRONMENT: Operating: 0° to 50°C up to 35°C at 70% R.H. **Storage:** -25° to 65°C .



6522 SPECIFICATIONS

CHANNELS PER CARD: 10.
FUNCTIONS: Volts, Amps.
CONTACT CONFIGURATION: Single pole, "break-before-make" for signal HI input. Signal LO is common for all 10 channels and output. When a channel is off, signal HI is connected to signal LO. 6517B can also configure channels as voltage switches.
CONNECTOR TYPE: Inputs: Triaxial. Outputs: Triaxial.
SIGNAL LEVEL: 200V, 500mA, 10VA (resistive load).
CONTACT LIFE: $>10^6$ closures at maximum signal level; $>10^7$ closures at low signal levels.
CONTACT RESISTANCE: $<1\Omega$.
CONTACT POTENTIAL: $<200\mu\text{V}$.
OFFSET CURRENT: $<1\text{pA}$ ($<30\text{fA}$ typical at 23°C , $<60\%$ RH).
CHANNEL ISOLATION: $>10^{13}\Omega$, $<0.3\text{pF}$.
INPUT ISOLATION: $>10^{10}\Omega$, $<125\text{pF}$ (Input HI to Input LO).
ACTUATION TIME: 2ms.
COMMON MODE VOLTAGE: $<300\text{V}$ peak.
ENVIRONMENT: Operating: 0° to 50°C up to 35°C at 70% R.H. **Storage:** -25° to 65°C .



VOLTS

RANGE	6½-DIGIT RESOLUTION	ACCURACY (1 Year) ¹ 18°–28°C ±(% + offset)	TEMPERATURE COEFFICIENT 0°–18°C & 28°–50°C ±(% + offset)/°C
2 V	1 µV	0.025 + 40 µV	0.003 + 20 µV
20 V	10 µV	0.025 + 300 µV	0.002 + 100 µV
200 V	100 µV	0.06 + 3 mV	0.002 + 1 mV

NMRR: 2V and 20V ranges >60dB, 200V range >55dB. 50Hz or 60Hz².

CMRR: >120dB at DC, 50Hz or 60Hz.

INPUT IMPEDANCE: >200TΩ in parallel with 20pF, <2pF guarded (1MΩ with zero check on).

SMALL SIGNAL BANDWIDTH AT PREAMP OUTPUT: Typically 100kHz (–3dB).

NOTES

- When properly zeroed, 6½-digit, 1 PLC (power line cycle), median filter on, digital filter = 10 readings.
- Line sync on.

AMPS

RANGE	6½-DIGIT RESOLUTION	ACCURACY (1 Year) ¹ 18°–28°C ±(% + offset)	TEMPERATURE COEFFICIENT 0°–18°C & 28°–50°C ±(% + offset)/°C
20 pA	10 aA ²	1 + 3 fA	0.1 + 500 aA
200 pA	100 aA ²	1 + 5 fA	0.1 + 1 fA
2 nA	1 fA	0.2 + 300 fA	0.1 + 20 fA
20 nA	10 fA	0.2 + 500 fA	0.03 + 100 fA
200 nA	100 fA	0.2 + 5 pA	0.03 + 1 pA
2 µA	1 pA	0.1 + 100 pA	0.005 + 20 pA
20 µA	10 pA	0.1 + 500 pA	0.005 + 100 pA
200 µA	100 pA	0.1 + 5 nA	0.005 + 1 nA
2 mA	1 nA	0.1 + 100 nA	0.008 + 20 nA
20 mA	10 nA	0.1 + 500 nA	0.008 + 100 nA

INPUT BIAS CURRENT: <3fA at T_{CAL}. Temperature coefficient = 0.5fA/°C, 20pA range.

INPUT BIAS CURRENT NOISE: <750aA p-p (capped input), 0.1Hz to 10Hz bandwidth, damping on. Digital filter = 40 readings, 20pA range.

INPUT VOLTAGE BURDEN at T_{CAL} ±1°C:

<20µV on 20pA, 2nA, 20nA, 2µA, and 20µA ranges.

<100µV on 200pA, 200nA, and 200µA ranges.

<2mV on 2mA range. <5mV on 20mA range.

TEMPERATURE COEFFICIENT OF INPUT VOLTAGE BURDEN: <10µV/°C on pA, nA, and µA ranges.

PREAMP SETTLING TIME (to 10% of final value) Typical: 0.5sec (damping off)

2.0 sec (damping on) on pA ranges. 15msec on nA ranges damping off, 1msec on µA ranges damping off. 500µsec on mA ranges damping off.

NMRR: >60dB on all ranges at 50Hz or 60Hz².

NOTES

- When properly zeroed, 6½-digit, 1PLC (power line cycle), median filter on, digital filter = 10 readings.
- aA = 10^{–18}A, fA = 10^{–15}A.
- Line sync on.

OHMS (Normal Method)

RANGE	6½-DIGIT RESOLUTION	ACCURACY ¹ (10–100% Range) 18°–28°C (1 Year) ±(% + offset)	TEMPERATURE COEFFICIENT (10–100% Range) 0°–18°C & 28°–50°C ±(% + offset)	AUTO V SOURCE	AMPS RANGE
2 MΩ	1 Ω	0.125 + 10 Ω	0.01 + 10 Ω	40 V	200 µA
20 MΩ	10 Ω	0.125 + 100 Ω	0.01 + 100 Ω	40 V	20 µA
200 MΩ	100 Ω	0.15 + 1 kΩ	0.015 + 1 kΩ	40 V	2 µA
2 GΩ	1 kΩ	0.225 + 10 kΩ	0.035 + 10 kΩ	40 V	200 nA
20 GΩ	10 kΩ	0.225 + 100 kΩ	0.035 + 100 kΩ	40 V	20 nA
200 GΩ	100 kΩ	0.35 + 1 MΩ	0.110 + 1 MΩ	40 V	2 nA
2 TΩ	1 MΩ	0.35 + 10 MΩ	0.110 + 10 MΩ	400 V	2 nA
20 TΩ	10 MΩ	1.025 + 100 MΩ	0.105 + 100 MΩ	400 V	200 pA
200 TΩ	100 MΩ	1.15 + 1 GΩ	0.125 + 1 GΩ	400 V	20 pA

NOTES

- Specifications are for auto V-source ohms, when properly zeroed, 6½-digit, 1PLC, median filter on, digital filter = 10 readings. If user selectable voltage is required, use manual mode. Manual mode displays resistance (up to 10¹⁸Ω) calculated from measured current. Accuracy is equal to accuracy of V-source plus accuracy of selected Amps range.

PREAMP SETTLING TIME: Add voltage source settling time to preamp settling time in Amps specification. Ranges over 20GΩ require additional settling based on the characteristics of the load.

OHMS (ALTERNATING POLARITY METHOD)

The alternating polarity sequence compensates for the background (offset) currents of the material or device under test. Maximum tolerable offset up to full scale of the current range used.

Using Keithley 8009 fixture

REPEATABILITY: $\Delta I_{BG} \times R/V_{ALT} + 0.1\%$ (1σ) (instrument temperature constant ±1°C).

ACCURACY: $(V_{SRC}Err + I_{MEAS}Err \times R)/V_{ALT}$

where: ΔI_{BG} is a measured, typical background current noise from the sample and fixture.

V_{ALT} is the alternating polarity voltage used.

$V_{SRC}Err$ is the accuracy (in volts) of the voltage source using V_{ALT} as the setting.

$I_{MEAS}Err$ is the accuracy (in amps) of the ammeter using V_{ALT}/R as the reading.

VOLTAGE SOURCE

RANGE	5½-DIGIT RESOLUTION	ACCURACY (1 Year) 18°–28°C ±(% setting + offset)	TEMPERATURE COEFFICIENT 0°–18°C & 28°–50°C ±(% setting+offset)/°C
100 V	5 mV	0.15 + 10 mV	0.005 + 1 mV
1000 V	50 mV	0.15 + 100 mV	0.005 + 10 mV

MAXIMUM OUTPUT CURRENT:

100V Range: ±10mA, hardware short circuit protection at <14mA.

1000V Range: ±1mA, hardware short circuit protection at <1.4mA.

SETTLING TIME:

100V Range: <8ms to rated accuracy.

1000V Range: <50ms to rated accuracy.

NOISE (typical):

100V Range: <2.6mV rms.

1000V Range: <2.9mV rms.

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COULOMBS

RANGE	6½-DIGIT RESOLUTION	ACCURACY (1 Year) ^{1,2} 18°–28°C	TEMPERATURE COEFFICIENT
		±(% + offset)	0°–18°C & 28°–50°C ±(% + offset)/°C
2 nC	1 fC	0.4 + 50 fC	0.04 + 30 fC
20 nC	10 fC	0.4 + 500 fC	0.04 + 100 fC
200 nC	100 fC	0.4 + 5 pC	0.04 + 1 pC
2 µC	1 pC	0.4 + 50 pC	0.04 + 10 pC

NOTES

1. Specifications apply immediately after charge acquisition. Add $(4fA + \frac{|Q_{av}|}{RC}) T_A$ where T_A = period of time in seconds between the coulombs zero and measurement and Q_{av} = average charge measured over T_A , and $RC = 300,000$ typical.
2. When properly zeroed, 6½-digit, 1PLC (power line cycle), median filter on, digital filter = 10 readings.
- INPUT BIAS CURRENT: <4fA at T_{CAL} . Temperature coefficient = 0.5fA/°C, 2nC range.

TEMPERATURE (Thermocouple)

THERMOCOUPLE TYPE	RANGE	ACCURACY (1 Year) ¹ 18°–28°C
		±(% rdg + °C)
K	–25°C to 150°C	±(0.3% + 1.5°C)

NOTES

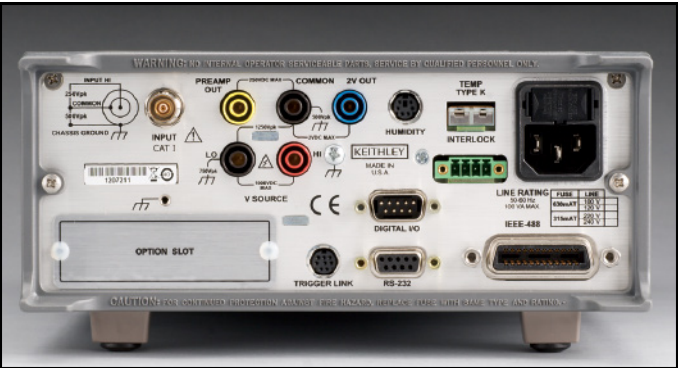
1. Excluding probe errors, $T_{cal} \pm 5^\circ\text{C}$, 1 PLC integration time.

HUMIDITY

RANGE	ACCURACY (1 Year) ¹ 18°–28°C, ±(% rdg + % RH)
0–100%	±(0.3% + 0.5)

NOTES

1. Humidity probe accuracy must be added. This is ±3% RH for Model 6517-RH, up to 65°C probe environment, not to exceed 85°C.



6517B rear panel.

IEEE-488 BUS IMPLEMENTATION

IMPLEMENTATION: SCPI (IEEE-488.2, SCPI-1999.0).
TRIGGER TO READING DONE: 150ms typical, with external trigger.
RS-232 IMPLEMENTATION: Supports: SCPI 1991.0. Baud Rates: 300, 600, 1200, 2400, 4800, 9600, 19.2k, 38.4k, 57.6k, and 115.2k.
FLOW CONTROL: None, Xon/Xoff.
CONNECTOR: DB-9 TXD/RXD/GND.

GENERAL

OVERRRANGE INDICATION: Display reads “OVERFLOW” for readings >105% of range. The display reads “OUT OF LIMIT” for excessive overrange conditions.
RANGING: Automatic or manual.
CONVERSION TIME: Selectable 0.01PLC to 10PLC.
MAXIMUM INPUT: 250V peak, DC to 60Hz sine wave; 10sec per minute maximum on mA ranges.
MAXIMUM COMMON MODE VOLTAGE (DC to 60Hz sine wave): Electrometer, 500V peak; V Source, 750V peak.
ISOLATION (Meter COMMON to chassis): >10¹⁰Ω, <500pF.
INPUT CONNECTOR: Three lug triaxial on rear panel.
2V ANALOG OUTPUT: 2V for full range input. Non-inverting in Volts mode, inverting when measuring Amps, Ohms, or Coulombs. Output impedance 10kΩ.
PREAMP OUTPUT: Provides a guard output for Volts measurements. Can be used as an inverting output or with external feedback in Amps and Coulombs modes.
EXTERNAL TRIGGER: TTL compatible External Trigger and Electrometer Complete.
GUARD: Switchable voltage guard available.
DIGITAL I/O AND TRIGGER LINE: Available, see manual for usage.
EMC: Conforms to European Union Directive 89/336/EEC, EN 61326-1.
SAFETY: Conforms to European Union Directive 73/23/EEC, EN 61010-1.
READING STORAGE: 50,000.
READING RATES:
 To Internal Buffer: 425 readings/second¹.
 To IEEE-488 Bus: 400 readings/second^{1,2}.
 Bus Transfer: 3300 readings/second².
 1. 0.01PLC, digital filters off, front panel off, temperature + RH off, Line Sync off.
 2. Binary transfer mode.
DIGITAL FILTER: Median and averaging.
ENVIRONMENT: Operating: 0°–50°C; relative humidity 70% non-condensing, up to 35°C.
Storage: –25° to +65°C.
ALTITUDE: Maximum 2000 meters above sea level per EN 61010-1.
WARM-UP: 1 hour to rated accuracy (see manual for recommended procedure).
POWER: User selectable 100, 120, 220, 240VAC ±10%; 50/60Hz, 100VA max.
PHYSICAL: Case Dimensions: 90mm high × 214mm wide × 369mm deep (3½ in. × 8½ in. × 14½ in.).
Working Dimensions: From front of case to rear including power cord and IEEE-488 connector: 15.5 inches.
Net Weight: 5.4kg (11.8 lbs.).
Shipping Weight: 6.9kg (15.11 lbs.).

Contact Information:

ASEAN / Australia (65) 6356 3900
Austria 00800 2255 4835
Balkans, Israel, South Africa and other ISE Countries +41 52 675 3777
Belgium 00800 2255 4835
Brazil +55 (11) 3759 7627
Canada 1 800 833 9200
Central East Europe and the Baltics +41 52 675 3777
Central Europe & Greece +41 52 675 3777
Denmark +45 80 88 1401
Finland +41 52 675 3777
France 00800 2255 4835
Germany 00800 2255 4835
Hong Kong 400 820 5835
India 000 800 650 1835
Italy 00800 2255 4835
Japan 81 (3) 6714 3010
Luxembourg +41 52 675 3777
Mexico, Central/South America & Caribbean 52 (55) 56 04 50 90
Middle East, Asia, and North Africa +41 52 675 3777
The Netherlands 00800 2255 4835
Norway 800 16098
People's Republic of China 400 820 5835
Poland +41 52 675 3777
Portugal 80 08 12370
Republic of Korea 001 800 8255 2835
Russia & CIS +7 (495) 6647564
South Africa +41 52 675 3777
Spain 00800 2255 4835
Sweden 00800 2255 4835
Switzerland 00800 2255 4835
Taiwan 886 (2) 2656 6688
United Kingdom & Ireland 00800 2255 4835
USA 1 800 833 9200

Rev 0415

For Further Information

Tektronix maintains a comprehensive, constantly expanding collection of application notes, technical briefs and other resources to help engineers working on the cutting edge of technology. Visit www.tektronix.com or www.keithley.com.

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